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A brief overview of diabetes detection using permutation entropy

1. Introduction

1.1. Materials

For this project, we are using the following materials:

- DBT2 and NDBT2, 2 datasets of 20 diabetics and non diabetics patients each. The .mat files are composed of the glucose concentration **G** vs time (T_{sample} = 15min ie F_{sample} = 1/900 Hz) and the "food activity" **S** vs time, linked to the alimentation of each patient.
- The permutation entropy and all its derivate concepts. We will use MATLAB and the EntropyHUB library as well as homemade code made by Dr. Walter Legnani and its students.

1.2. Goals and plan

The main goal of this project is to develop a classifier that is able to distinguish between diabetics and non diabetics patients. To achieve this, we will try several machine learnings techniques based on well-chosen metrics and parameters.

Rather than analysing the glucose concentration alone, we also try to correlate it with the "food activity" as mentionned earlier, in order to develop a more robust classifier that is able to take into account the alimentation of the patients.

We can build the plan of the project as follows:

1. A preliminary study to find the most relevant embedding dimension (**m**) and embedding time delay ($\boldsymbol{\tau}$) in order to apply permutation entropy tools
2. The selection of the most relevant metrics and parameters to feed the classifier
3. The choice and development of the classifier
4. The evaluation of the classifier's performance and its optimization

2. Preliminary study: finding the most relevant (m,τ) combination

2.1. CxH Feature plan

To find the most relevant embedding dimension (**m**) and embedding time delay ($\boldsymbol{\tau}$), we first focused on the evolution of glucose's concentration over time. We used the standard Complexity-Entropy feature plan in order to find the combination that splits the most the two classes of patients. To be more precise, we used the MPR Complexity denoted C and the normalized permutation entropy denoted H.

2.2. Finding the parameters

To compare the two classes, we computed the euclidean centroid of each class and then the euclidean distance between the two centroids. The combination of (m,τ) that maximizes this distance is the one we will use to compute the metrics for the classifier.

We know that we consider more or less chaotic behaviors so to give an order of magnitude, we expect the centroids to have a entropy H < 0.9 and a complexity C > 0.1.

In order to find bounds to the value of the parameters, the litterature suggests that usually $m \in [3, 7]$ and rarely exceeds 5 or 6 for physiological signals.

We also chose to limit the maximum value of τ to 3, because we know that the glucose concentration is sampled every 15 minutes, so a τ of 3 would correspond to a time delay of 45 minutes, which is already quite long for a physiological signal.

Using the MATLAB functions `HXC_centroid.m` and `HXC_param_sweep_euclide.m`, we computed the distance between the two centroids for all the combinations :

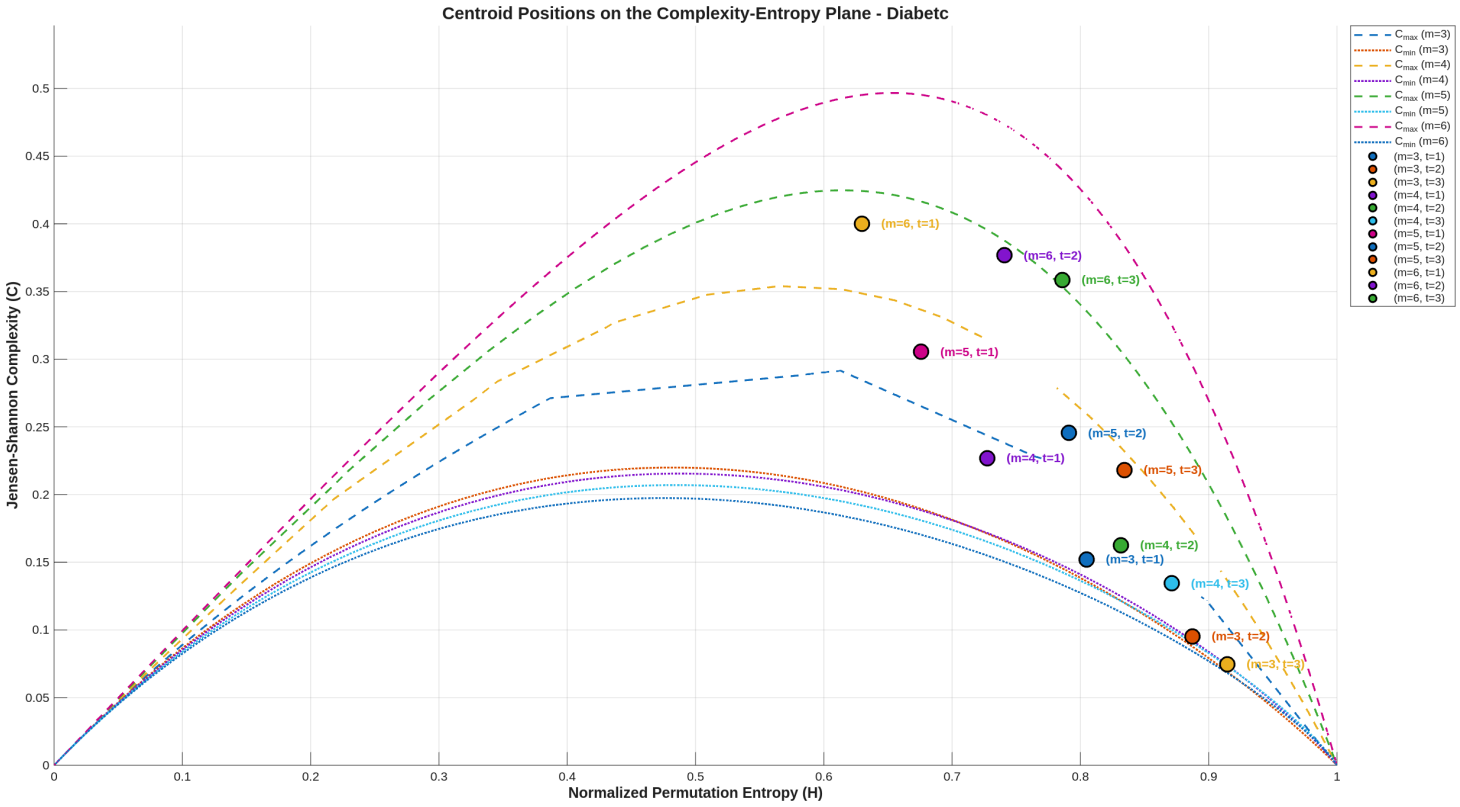


Figure 1: HxC feature plan of the centroids of the diabetic dataset

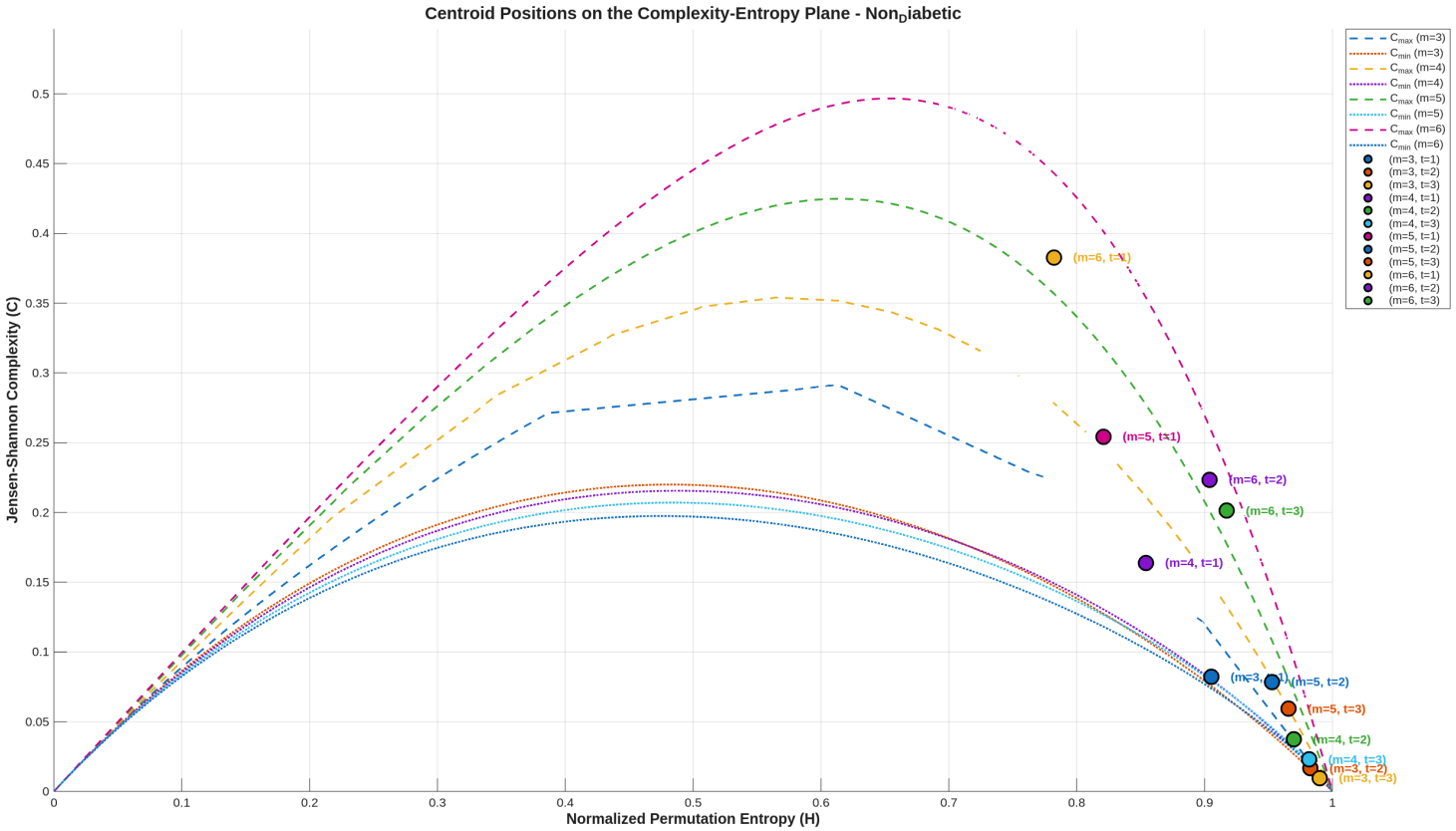


Figure 2: HxC feature plan of the centroids of the non-diabetic dataset

At first glance, we have several candidats that behave a physilogical chaotic signal. We obtain after computation :

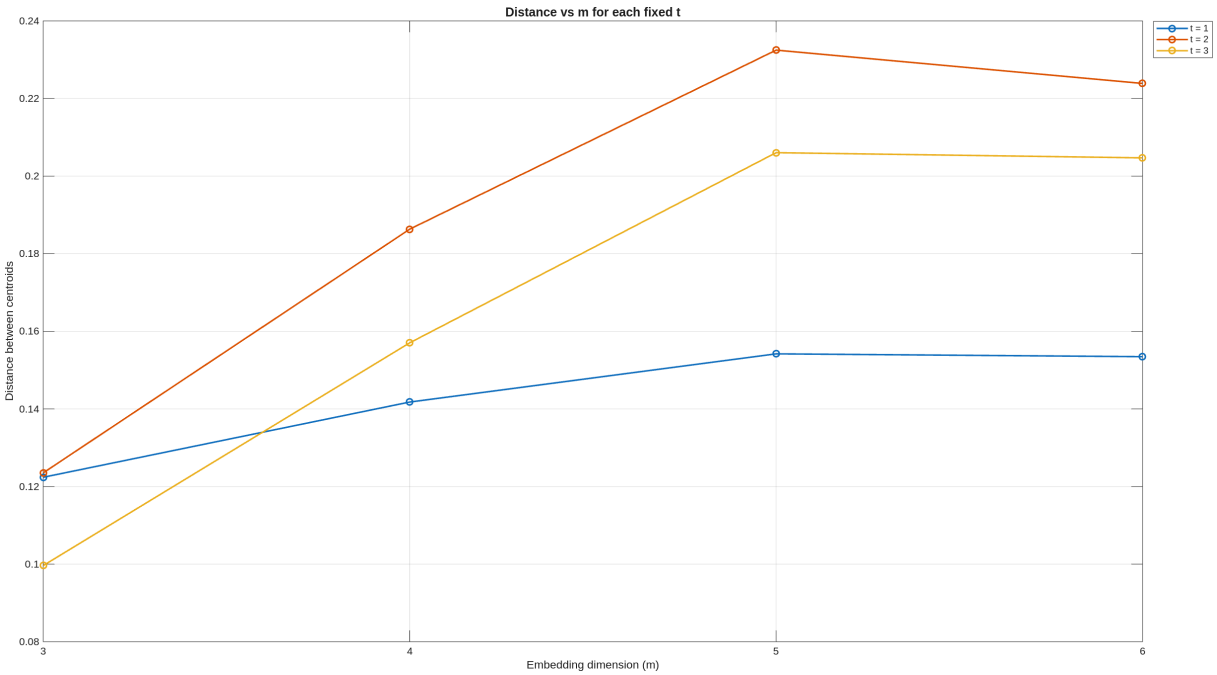


Figure 3: Distance between centroids vs. embedding dimension and time delay

We can furthermore compute the standard deviation around the centroids to be sure they do not overlap:

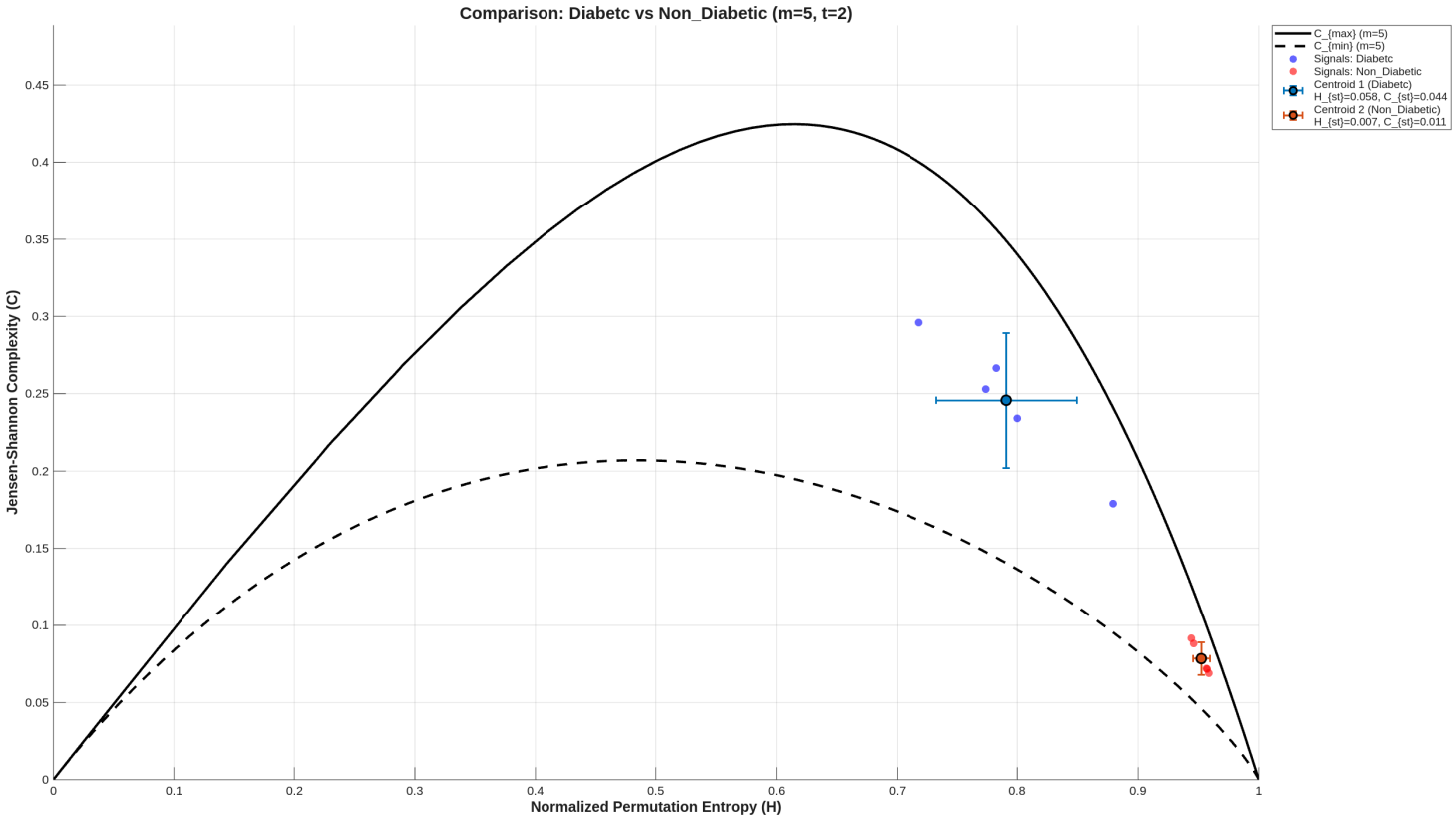


Figure 4: HxC feature plan with standard deviation of the (5,2) combination

Result : From this approach, we concluded that $\mathbf{(m = 5, \tau = 2)}$ is the most relevant combination to compute permutation entropy.

NB : This way of selection is mainly empirical and does not necesseraly provide the optimized combination. To be more precise on the choice of τ, we can also compute the autocorrelation of the datas.

3. Selections of the most relevant metrics

Now that we have found a satisfying set of parameters, we have to find metrics that link **G** and **S** and show different behaviors between diabetic and non-diabetic patients.

3.1. Mutual information

To have an idea of the quantity of information shared between **G** and **S**, we computed with `internal_PMI_sweep.m` the mutual information of each patient and the mean mutual information of each class :

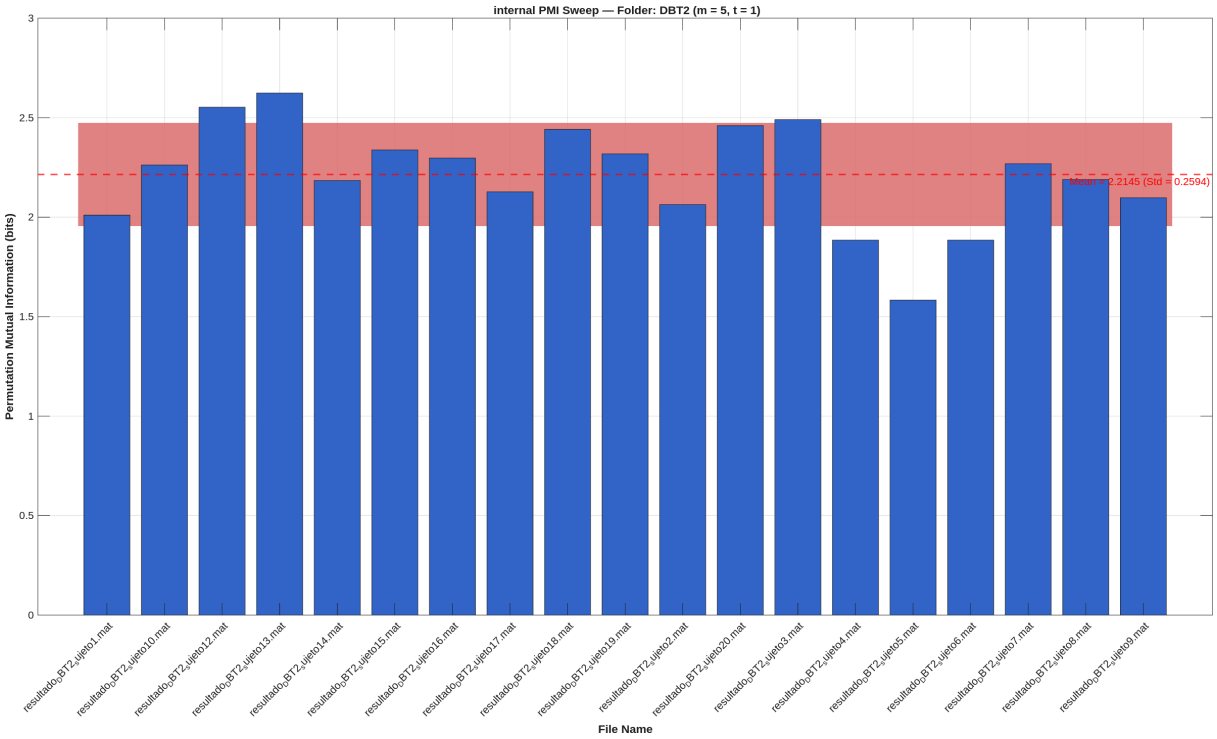


Figure 5: Mean, standard deviation and individual mutual information of G and S for each diabetic patient

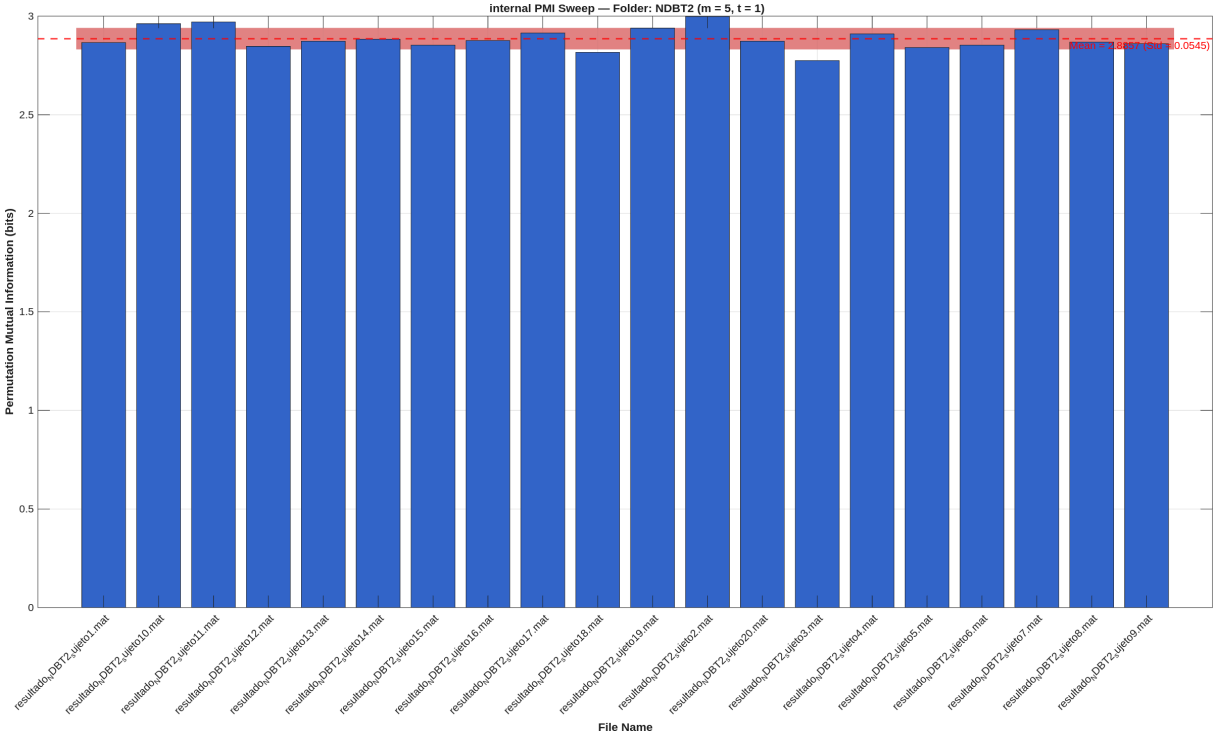


Figure 6: Mean, standard deviation and individual mutual information of G and S for each non-diabetic patient

Group of patient	Mean Mutual Information of G and S	Standard deviation
Diabetic	2.2145	0.2594
Non-diabetic	2.8857	0.0545

Table 1: Mean mutual information of G and S for each class of patients

Result : The mean mutual information of the non-diabetic patients is higher than the one of the diabetic patients, which means that the glucose concentration is more correlated to the food activity for healthy patients.

We observe as well that diabetic patients have more spreaded mutual information values, meaning one more time as smaller correlation between the glucose concentration and the food activity for them.

3.2. Transfer entropy

To go further on the characterisation of the correlation between **G** and **S**, we computed the transfer permutation entropy in order to measure the directed transfer of information between the two signals **G** and **S**, using the `internal_TE_sweep.m` function :

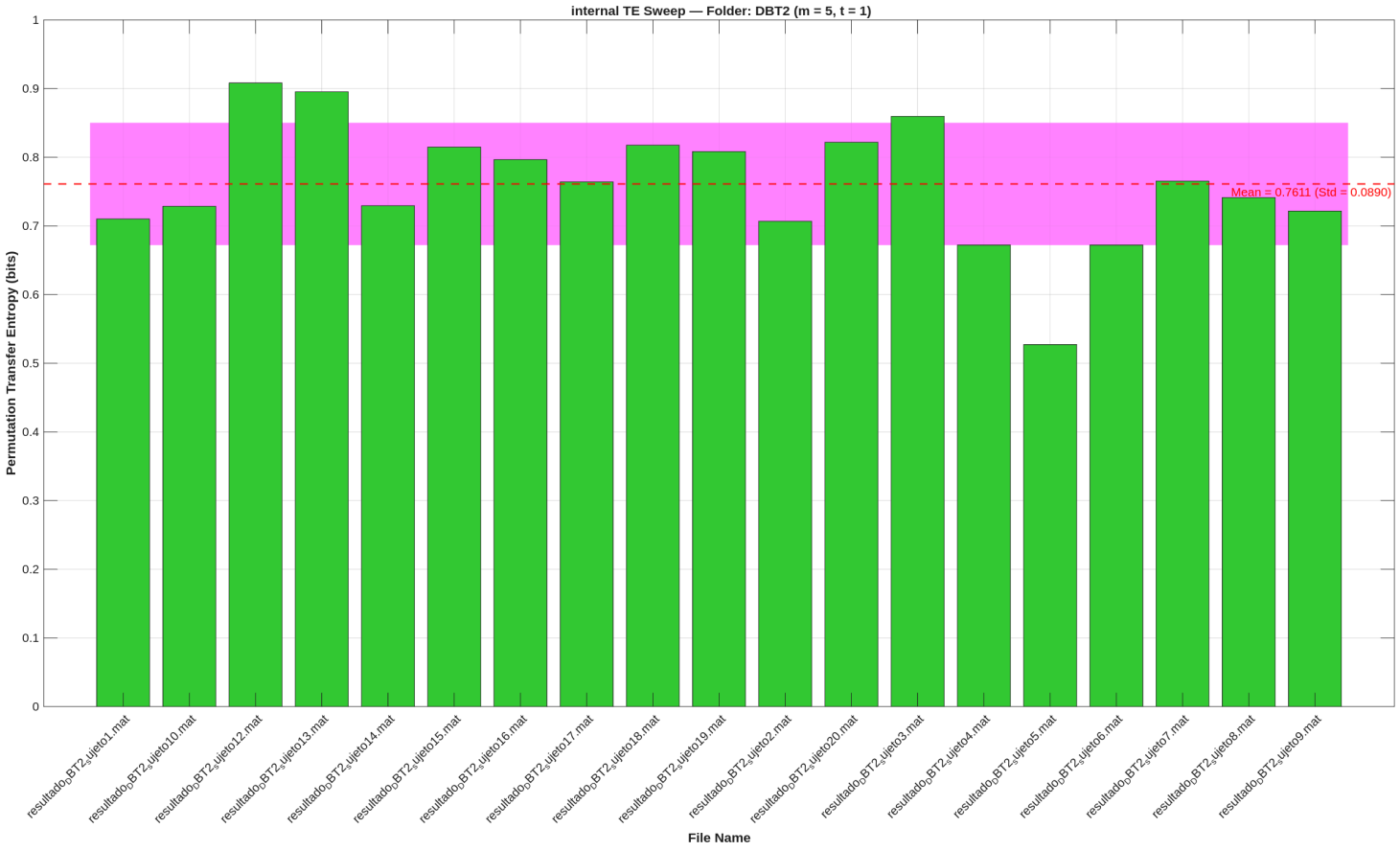


Figure 7: Mean, standard deviation and individual transfer entropy of G and S for each diabetic patient

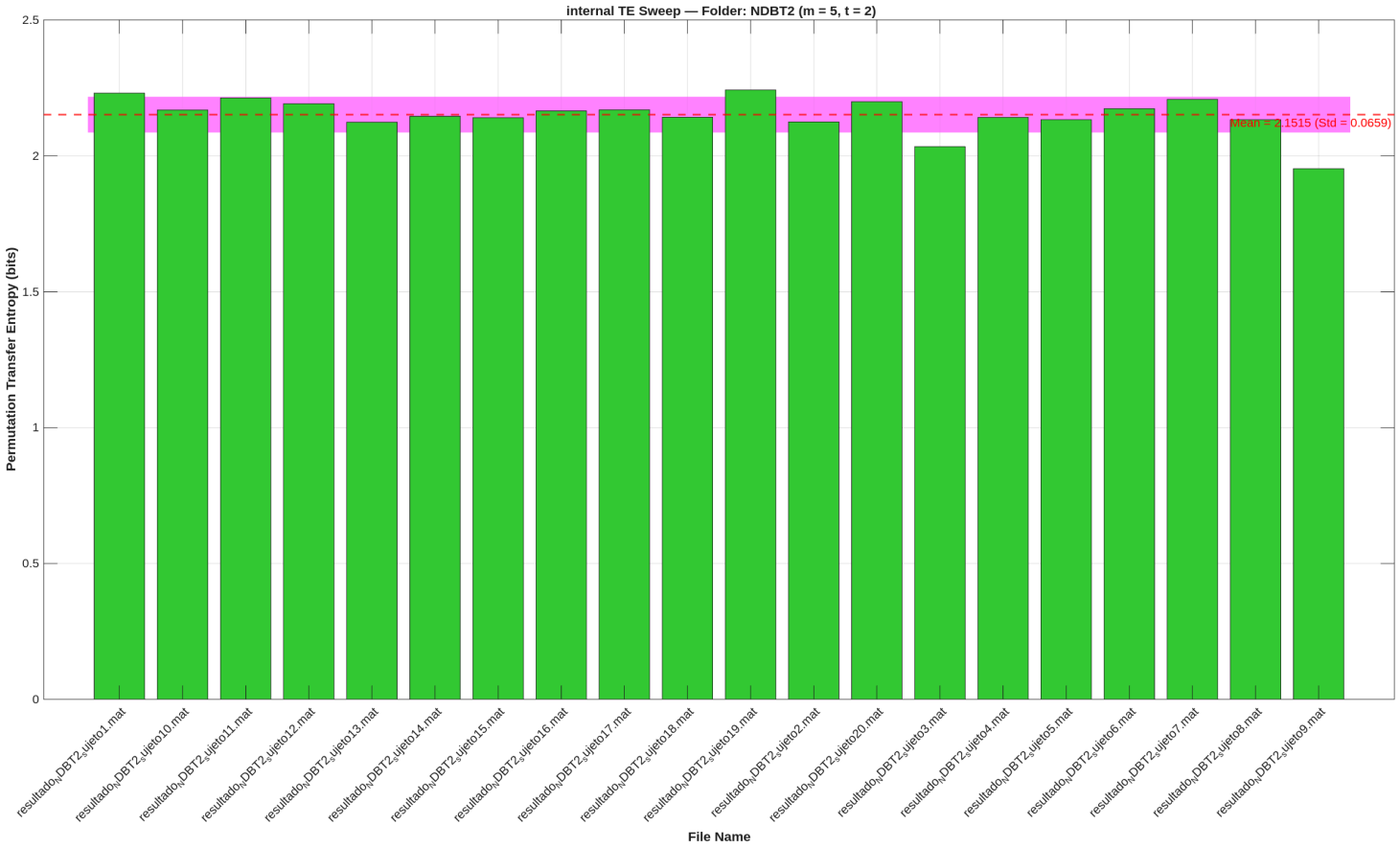


Figure 8: Mean, standard deviation and individual transfer entropy of G and S for each non-diabetic patient

Group of patient	Mean Transfer Permutation Entropy S → G	Standard deviation
Diabetic	0.7611	0.0890
Non-diabetic	2.1515	0.0659

Table 2: Mean mutual information of G and S for each class of patients

Result : The transfer permutation entropy of non-diabetic patient is around 3 times higher than the one of diabetic patients, highlighting a stronger dependency of **G** on **S** for healthy patients. This is consistent with the mutual information results.

Once again, the diabetic patients values are more spreaded, underlining a thiner correlation between glucose concentration and food activity for them.

To be continued...

4. To try next

4.1. Multi-dimensional

- 2-D Permutation Entropy (see EntropyHUB)

4.2. Mono-dimensional

On the glucose concentration ?

- Fourier Spectrum
- AUC (power density)
- Fractal dimension (with Higuchi's method)
- Lyapunov exponent